

Computational workflow for the reproducibility datasets and figures

The same field and momentum conventions are used
by both independent mode formulations

1 Set parameters

Pulse parameters and units
Momentum grid p/m
Tail extent L/τ
Solver tolerances

2 Define the field

Sauter pulse or pulse train
 $E(t)$, $A(t)$, finite window

3 Solve each mode

Dirac spinor: DOP853
QKE state (f, u, v) : DOP853
Independent integrations

4 Validate each run

Exact Sauter benchmark
Dirac norm; QKE invariant
Dirac-QKE agreement;
tail test

5 Save datasets

Spectrum and yield CSV
Tail and p - L/τ map CSV
Run metadata JSON

6 Generate and build

Python/Matplotlib figures
LaTeX $\rightarrow$ two-pass CAS PDF
Package checksums

Scripts: `reproduce.py` · `make_reproducibility_figures.py`
`build_manuscript.sh`